

CRE Metric	How to Measure in Lab	Desired Target
Soot Oxidation Window	TGA/DSC of soot-coated cloth vs. bare soot	Peak oxidation shifted from 600°C → < 350°C
Durability / Mechanical Adhesion	Ultrasonic bath weight loss test in ethanol for 15 min	Catalyst peeling loss < 2 wt%

Sources

1. [Mars-van Krevelen Mechanism | Inorganic Chem II - Fiveable](#)
-

User prompt: can you write down in more detail from start as propsal with proper fluid diagrams and everything

Response: Lab Project Proposal: Chemical Reaction Engineering

Project Title: Development and Hydrodynamic/Kinetic Evaluation of a Catalytic Ceramic Filter Membrane for Simultaneous Particulate Filtration and Flue-Gas De-Pollution (CO/VOC Oxidation) in Kiln Exhaust

Course: Chemical Reaction Engineering Laboratory

Focus Areas: Heterogeneous Catalysis, Non-Ideal Porous Flow, Process Intensification, Local Environmental Abatement (Brick Kiln Smog)

1. Executive Summary & Problem Context

Traditional Bull’s Trench Kilns (BTKs) and modified Zig-Zag kilns across South Asia burn low-grade coal, scrap tires, and agricultural residues. The resulting exhaust contains hazardous levels of black carbon/fly ash, carbon monoxide (CO), unburned hydrocarbons/volatile organic compounds (VOCs), and sulfur/nitrogen oxides (SO_x/NO_x).

Current remediation approaches decouple particulate removal (cyclones, settling chambers) from gaseous pollutant scrubbers. Standard fixed-bed adsorption systems fail almost immediately due to fly-ash blinding, excessive pressure drops, and rapid sorbent saturation.

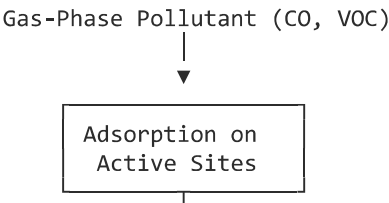
This project details the fabrication and bench-scale evaluation of an **integrated catalytic filter membrane**. High-temperature woven E-glass/ceramic fabric is functionalized via sol-gel dip-coating with a nanostructured mixed-metal oxide (CeO₂ – MnO_x). The material serves a dual function:

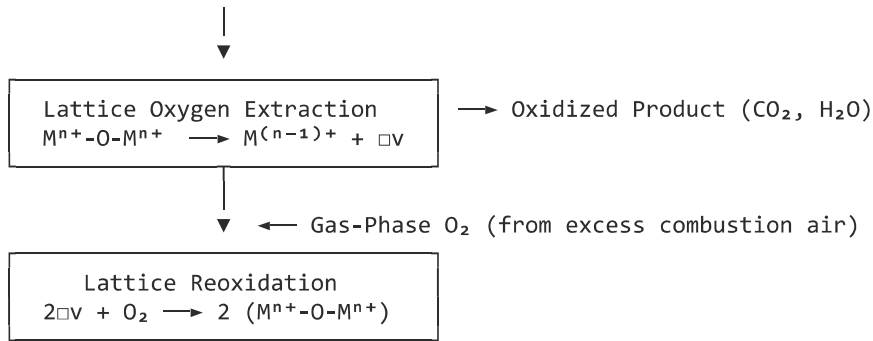
1. **Physical Surface Filtration:** Trapping particulate matter and soot on the exterior boundary face as a dynamic, removable filter cake.
2. **Heterogeneous Catalytic Oxidation:** Decomposing toxic gaseous species (CO and model VOCs) into inert products (CO₂, H₂O) upon trans-membrane pore diffusion via the Mars–van Krevelen mechanism.

2. Chemical Mechanisms & Kinetic Framework

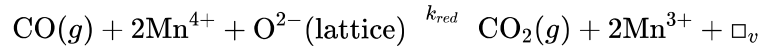
2.1 The Mars–van Krevelen (MvK) Redox Cycle

Unlike standard physical sorbents that saturate, transition metal redox catalysts (Ce⁴⁺/Ce³⁺ and Mn⁴⁺/Mn³⁺) act as continuous lattice-oxygen donors.





- **Surface Reduction Step:**



where $\square_v$ represents an anionic surface oxygen vacancy.

- **Surface Reoxidation Step:**



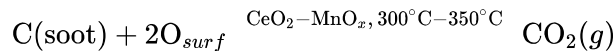
- **Overall Rate Law:** At steady state, equating the rates of reduction and oxidation yields the classic MvK rate expression:

$$-r_{CO} = \frac{k_{red}P_{CO} \cdot k_{ox}P_{O_2}}{\nu k_{red}P_{CO} + k_{ox}P_{O_2}}$$

where ν is the stoichiometric coefficient of oxygen consumption.

2.2 Low-Temperature Soot Self-Regeneration

Direct contact between unburned carbon soot and the catalytic coating lowers the soot activation energy:



This prevents tarry hydrocarbon binding on the fabric face.

3. Fluid Mechanics & Transport Phenomenon

The catalytic cloth functions as a thin porous flow-through wall reactor. Transport across the clean fabric and accumulating particulate cake is modeled via the **Forchheimer-extended Darcy Equation**:

$$-\frac{dP}{dz} = \frac{\mu}{\kappa} v_0 + \beta \rho v_0^2$$

where:

- v_0 = superficial gas face velocity (m/s)
- μ = flue-gas dynamic viscosity (Pa · s)
- κ = Darcy permeability coefficient of the clean membrane (m²)
- β = non-Darcian inertial form drag coefficient (m⁻¹)
- ρ = gas density (kg/m³)

Cake Resistance and Pressure Drop Dynamics

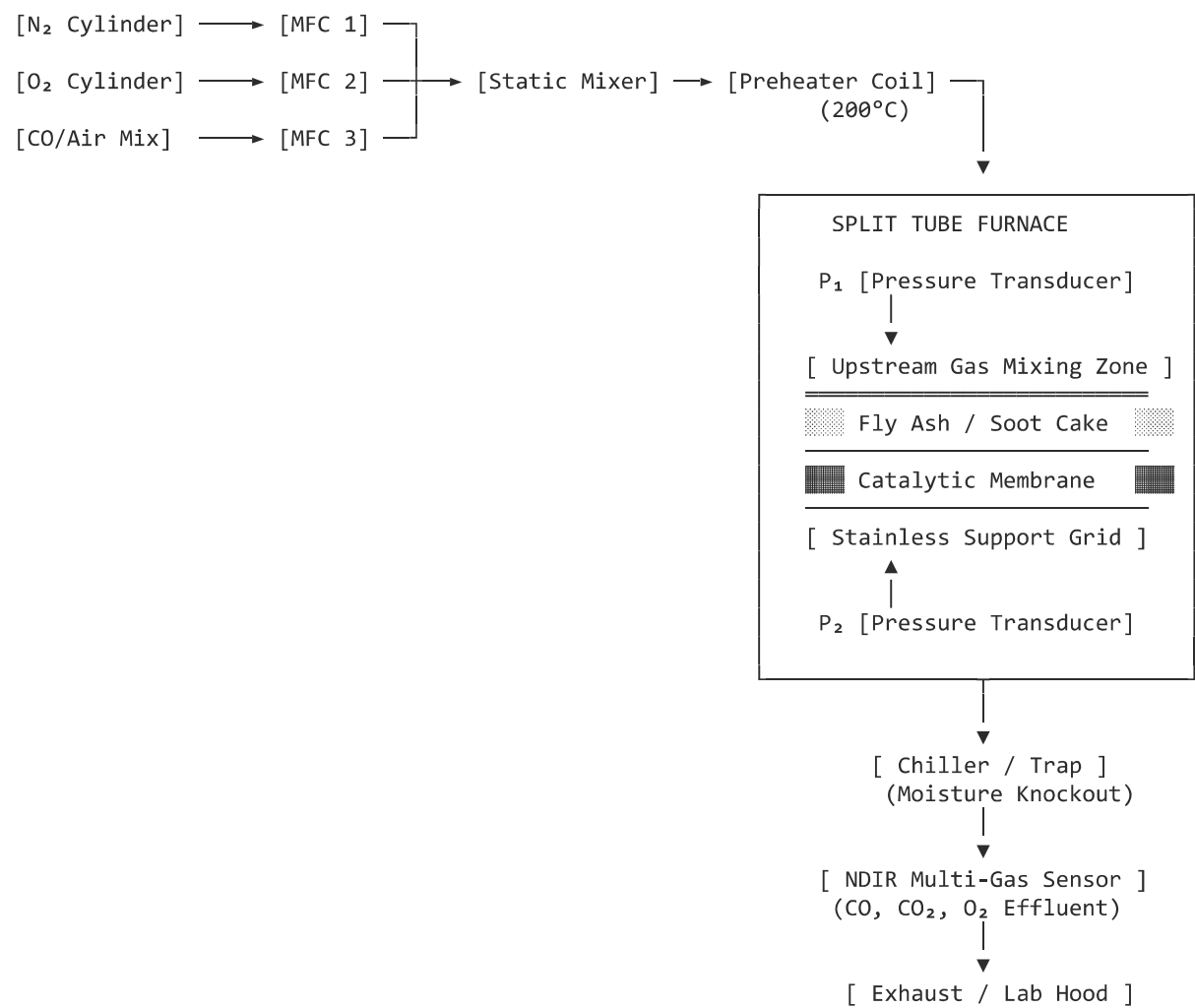
As soot and ash deposit on the upstream surface over time t , total pressure drop across the filter system (ΔP_{total}) expands into two resistances in series:

$$\Delta P_{total} = \Delta P_{membrane} + \Delta P_{cake} = v_0 \left(\frac{\mu \cdot L_m}{\kappa_m} + \mu \cdot r_c \cdot w_c(t) \right)$$

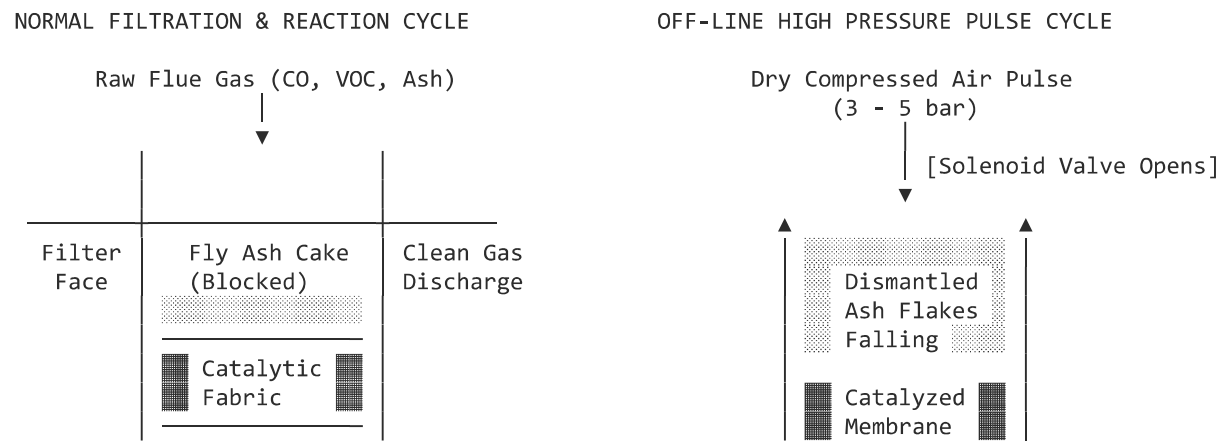
where L_m is membrane thickness (m), r_c is specific cake flow resistance (m/kg), and $w_c(t)$ is cake mass accumulated per unit area (kg/m²).

4. System Process Flow Diagrams (PFD)

4.1 Complete Laboratory Evaluation Skid (Gas Blending & Reaction System)



4.2 Cross-Flow / Reverse Pulse Cleaning Mechanism





5. Experimental Preparation Protocol

Raw E-Glass Cloth → [Desizing/Etch] → [Sol Synthesis] → [Dip-Coating] → [Controlled Calcination]

5.1 Substrate Pre-Treatment

- 1. Cut high-temperature woven E-glass or high-silica fabric into circular coupons (47 mm or 25 mm matching the reactor tube internal diameter).
- 2. Submerge in 10 wt% HNO₃ solution at 60°C for 45 minutes to chemically strip industrial silane binders and activate surface silanol groups (–Si-OH).
- 3. Rinse repeatedly with deionized water until neutral pH is achieved, dry at 105°C for 2 hours, and thermally clean at 400°C for 1 hour.

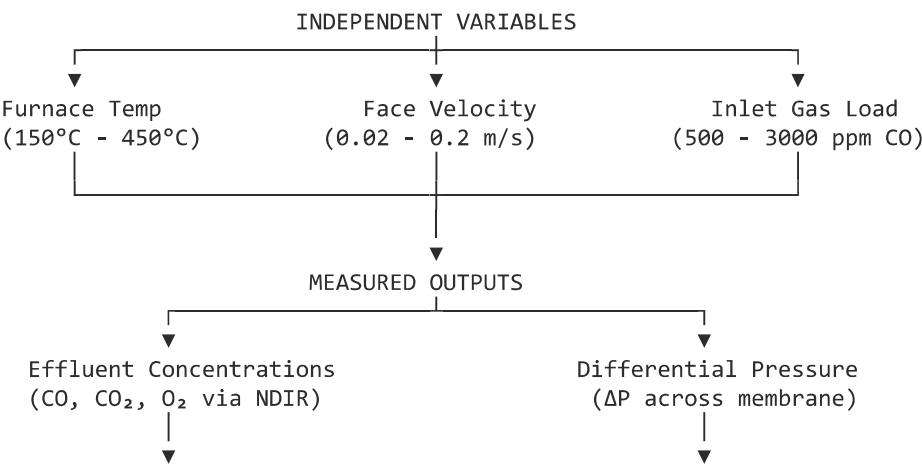
5.2 Catalytic Sol Preparation

- 1. Dissolve 0.05 mol of Ce(NO₃)₃ · 6H₂O and 0.05 mol of Mn(NO₃)₂ · 4H₂O in 100 mL of a 50/50 ethanol/ultrapure water mixture under magnetic stirring at 400 rpm.
- 2. Dissolve citric acid monohydrate (C₆H₈O₇ · H₂O) into the metal precursor solution at a 1.2 : 1 molar ratio of citric acid to total metal ions.
- 3. Heat solution to 65°C under reflux for 90 minutes until a transparent, viscous polymeric sol is obtained.

5.3 Deposition and Thermal Annealing

- 1. Mount the cleaned fabric coupons in a vertical dipping jig. Immerse into the sol for 5 minutes.
- 2. Withdraw vertically at a controlled rate of 1.5 cm/min to prevent gravity-induced meniscus pooling.
- 3. Pre-dry in a horizontal drying oven at 90°C for 1 hour.
- 4. Repeat immersion-drying sequence 3 times to achieve a uniform catalyst loading of 8 – 12 wt%.
- 5. Place dried samples into a programmable muffle furnace. Ramp at 2°C/min to 450°C in static air; dwell at 450°C for 3 hours to decompose all citrate/nitrate matrices into the crystalline CeO₂ – MnO_x phase.

6. Kinetic & Reactor Engineering Data Collection Matrix



Fractional Conversion
& Apparent Kinetics (Ea)

Permeability (κ) &
Cake Resistance (rc)

Primary Experimental Runs

Run Category	Primary Control Variable	Operating Range	Target CRE Calculation
I. Hydrodynamics	Face velocity (v_0)	0.01 – 0.25 m/s (Ambient)	Clean permeability (κ), Forchheimer parameter (β) via Darcy plot
II. Light-Off Test	Temperature (T)	150°C – 450°C (2°C/min ramp)	Conversion curves (X_{CO} vs T), T_{50} (temperature at 50% conversion)
III. Space Velocity	Gas Hourly Space Velocity	5,000 – 40,000 h ⁻¹ (adjusting flow)	Reaction order (n), intrinsic surface rate constant (k), Thiele modulus (ϕ)
IV. Deactivation	Time on Stream (TOS)	0 – 360 min continuous	Catalyst stability in presence of 200 ppm SO ₂ and moisture (5%)
V. Ash Deposition	Pre-loaded Carbon Black	0 – 50 g/m ² soot layer	Specific cake resistance (r_c), shift in soot ignition temp via differential pressure

7. Mathematical Formulations for the Final Report

1. Fractional Conversion (X_{CO}):

$$X_{CO} = \frac{C_{CO, in} - C_{CO, out}}{C_{CO, in}}$$

2. Apparent Activation Energy (E_a): Assuming low conversion differential reactor conditions ($X_{CO} < 0.15$):

$$-r'_{CO} = \frac{F_{CO, in} \cdot X_{CO}}{W_{cat}} = k_0 \exp\left(-\frac{E_a}{RT}\right) C_{CO}^n$$

Plot $\ln(-r'_{CO})$ vs. $1/T$ to determine E_a from the slope ($-E_a/R$).

3. Internal Diffusion Assessment (Weisz-Prater Criterion): To prove absence of pore-diffusion limitations within the catalyst-coated fiber layer:

$$C_{WP} = \frac{-r_{CO, obs} \cdot \rho_p \cdot R_p^2}{D_{eff} \cdot C_{CO, surf}} < 1$$

where R_p is effective fiber-coating thickness and D_{eff} is Knudsen/molecular binary diffusivity.

User prompt: see there is issue in docs of your flow diagram can make it exact what u wrote

Response: The issue in Google Docs happens because:

1. The text has a black background/highlight applied, making the text almost unreadable against the black fill.
2. Google Docs uses a **proportional font** (like Arial or Google Sans). ASCII art and box-drawing characters require a **monospaced font** (where every letter and space is the exact same width, like Courier New or Consolas) to stay aligned.

How to Fix It Instantly in Google Docs

1. Highlight the entire diagram in your Google Doc.